Covariate-artifact correction in epigenomic datasets

Mehdi Foroozandeh

Simon Fraser University.

Abstract

Epigenomic signal tracks carry artifacts from the way their reads were processed. We asked whether the recorded processing settings carry enough information for a model to remove these artifacts. We reprocessed the reads of seven ENCODE tracks with one setting changed at a time, which gave 130 versions of the same material with a known truth. We then trained small networks that read the source and target settings and transform one version into another. On setting changes seen in training, reading the settings lowered the error by a median fraction of 0.29, relative to a control trained with shuffled settings. On unseen combinations of settings, this fraction fell to 0.11, and no model reproduced the exact effect of sequencing depth. Larger networks, other losses, an added imputation objective and routing through the original processing did not close this gap. A mappability input helped in hard-to-map regions. Most results are exploratory and rest on seven tracks.

Keywords: epigenomics, imputation, batch effects, experimental covariates, counterfactual data, probabilistic forecasting

1 Motivation and background

Epigenome imputation predicts assays that were never run in a cell type from the assays that were. The ENCODE Imputation Challenge was a prospective benchmark of this task [1]. Its organisers collected the test data during the challenge, so no participant had seen them. Under the pre-registered measures, a baseline that averages each assay over the training cell types beat all but two of the 23 submitted models.

The organisers traced this failure to a processing difference. The training data were single-end reads and the test data were paired-end reads. The two run types used different duplicate-removal rules: a single-end read was kept once per duplicate set, while a read pair was kept if either mate was unique. This small difference shifted the

Table 1 The processing settings. Each reprocessed version differs from the original processing in one setting. The last four settings change only the p-value track, because they act on the control or on the peak caller.

Setting	Levels	What it changes
Sequencing depth	15M, 7.5M, 3.75M reads, as subsets of the original reads	counts and p-values
Read length	reads cropped to 36 or 50 bp before alignment	counts and p-values
Run type	paired-end instead of single-end	counts and p-values
Duplicate removal	turned off	counts and p-values
Mapping-quality filter	threshold lowered to 10 or 0	counts and p-values
Antibody-efficiency proxy	control reads mixed into the treatment to make up 50% or 90% of it	counts and p-values
Fragment extension	fragment length scaled by 0.5 or 2	p-values only
Control scaling	control signal scaled by 0.5 or 2	p-values only
Control identity	control from another cell type, or no control	p-values only
Control depth	control subsampled to 50% or 25%	p-values only

test signal enough to defeat most models. The authors concluded that such shifts need a correction that is more than a global rescaling.

Most imputation models ignore processing. CANDI, our imputation model, instead reads recorded experimental settings such as sequencing depth, run type and read length on both its input and its output. In principle, it can then predict what a track would look like under other settings. This ability cannot be tested inside CANDI's own training data. There, an experiment is paired only with randomly thinned copies of itself, so only depth ever changes while the material stays fixed. The other settings also change together across experiments, so their effects cannot be separated.

We therefore built a controlled testbed outside CANDI. We reprocessed the same reads with one setting changed at a time, so that the true effect of each change is known. We call the task of undoing these effects *covariate-artifact correction*. This paper asks one question: do the recorded processing settings carry enough information to map one processed version of a track onto another version of the same reads?

2 Methods

Data.

We took seven tracks from the challenge panel: one DNase-seq track, three histone marks with narrow peaks (H3K27ac, H3K4me1, H3K4me3) and three with broad domains (H3K27me3, H3K36me3, H3K9me3). We processed each track with the standard ENCODE pipeline at 30M reads (50M for DNase-seq). We call this the *original processing*. We then made *reprocessed versions*, each with exactly one processing setting changed (Table 1). Together with the originals, this gave 130 versions. Each version holds read counts in 25 bp bins and the peak caller's $-\log_{10} p$-value track. We also split every version into two random halves of its reads. The difference between the two halves measures sampling noise alone; we call it the *resampling floor*.

The correction model.

The model maps a source version X with settings C onto a target version X' with settings C' (Fig. 1a). It has two parts. A *conditioning network* reads C and C' and outputs parameters. A *transformation* applies these parameters to the source counts and outputs a negative binomial distribution for each 25 bp bin of the target. We tested four forms of the transformation, in increasing capacity: a *per-bin affine map* of the log counts; a *per-bin monotone curve*; a *smoothing kernel plus monotone curve*, which smooths over neighbouring bins first; and a *small convolutional network* (CNN) conditioned on the settings. Each form was trained either as one model per track or as one model shared across the seven tracks. In a second input variant, the conditioning network also read the source bin's own value. Four forms, two training scopes and two input variants give 16 configurations.

Trained and unseen setting changes.

We trained on pairs between the original processing and each reprocessed version, in both directions. We call these *trained setting changes*. We scored the models also on pairs between two reprocessed versions of the same track, which the models never saw in training (Fig. 1b). We call these *unseen setting changes*. An unseen change can move one setting between two levels, for example from 15M to 3.75M reads, or combine two settings, for example from 36 bp reads to paired-end. Models were trained on all chromosomes except chr19, chr21 and chr22; chr22 served for early stopping, and all scores are on chr19 and chr21.

Comparisons and controls.

The main control is a *shuffled-settings control*: the same model trained with the settings redrawn at random at each step, so that they carry no information. A gain over this control shows that the model uses the settings. The *covariate-free monotone baseline* fits one monotone map from source to target values over the whole genome, separately for each pair, without settings. For depth versions we also used an exact oracle, which thins the source reads binomially to the target depth. Depth also gives an exact rule: when depth changes by a factor r, the expected total count changes by r. We call this the *depth rule*. We measure its error as the median absolute $\log_2$ difference between the predicted and the true ratio of total counts.

Scoring.

We score predictions with the continuous ranked probability score (CRPS). CRPS measures the distance between a predicted distribution and the observed count; lower is better, and for a point forecast it equals the absolute error. We always split CRPS into two parts that add up to it. The *shape* part is the CRPS after the prediction is rescaled by the single best multiplicative factor for its track. The *level* part is what that factor removes. Every model ran with 3 random seeds (6 in one experiment). The *seed range* is the largest minus the smallest value over seeds, and we write a value as mean $\pm$ seed range. We call a difference between two models significant when it exceeds 2 times the larger of their two seed ranges.

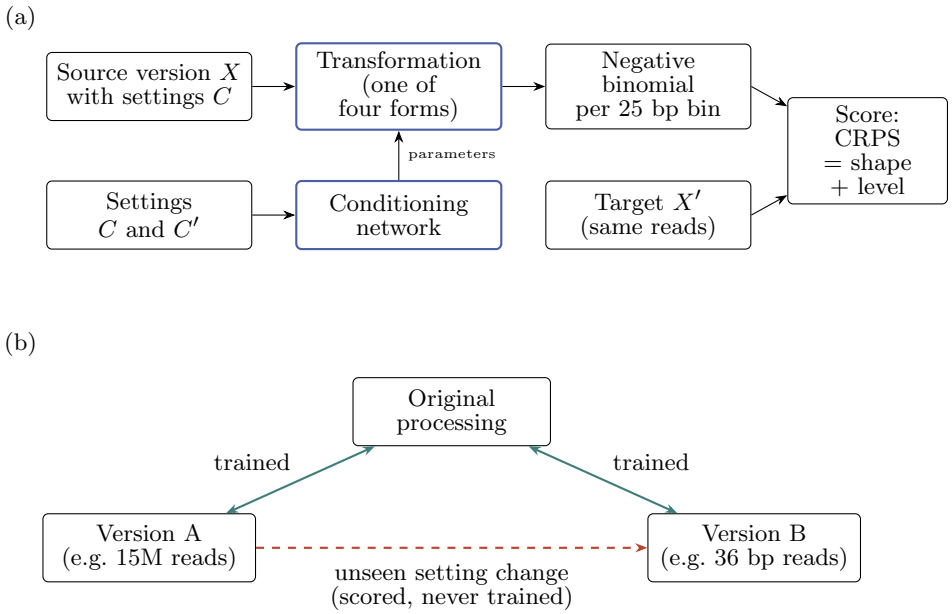


Fig. 1 Design. (a) The correction model. A conditioning network reads the source and target settings and sets the parameters of a transformation. The transformation turns the source counts into a negative binomial prediction of the target, which is scored by CRPS against the real target. (b) Training and scoring pairs within one track. Models train on pairs between the original processing and each reprocessed version (solid). They are scored also on pairs between two reprocessed versions (dashed), which they never saw.

Pre-registration.

Four experiments were pre-registered, that is, their checks and thresholds were fixed in writing before any run: training on pairs between reprocessed versions (Section 3.7), routing a shared model through the original processing (Section 3.8), an added imputation objective (Section 3.10) and a mappability input (Section 3.11). In the last three, a one-seed trial run exposed defects. We amended the design before the full runs: we added shuffled-settings controls, replaced a consistency loss that collapsed with an agreement loss between routes, and redefined empty mappability strata. Before the runs, the probability that every check would pass if the claim were true was 41% to 51%. All other results are exploratory: their thresholds were set after the data were seen.

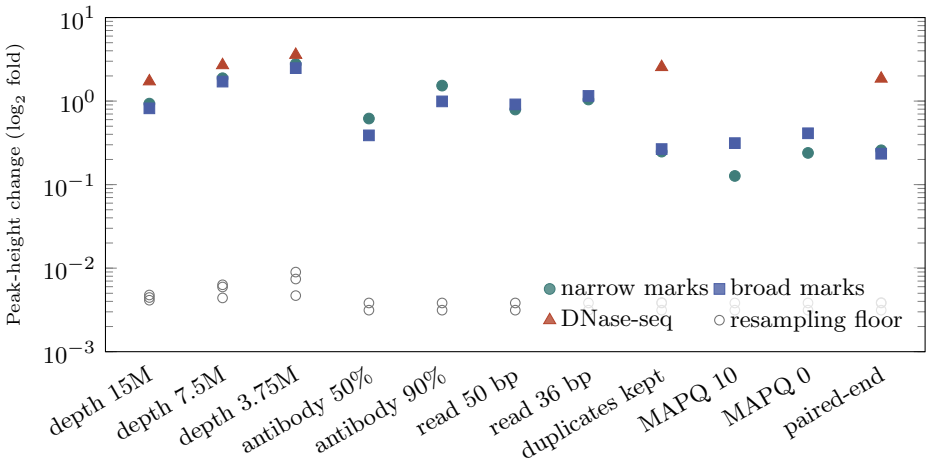


Fig. 2 Peak-height change caused by each setting level, against the resampling floor. Filled marks show the median change over the tracks of a mark class, on peaks shared by the original and the reprocessed version; DNase-seq has no version for some settings. Open grey circles show the floor, the change between two random halves of the same reads at the same depth. The y axis is logarithmic. There are no error bars: each value is deterministic, and the floor's seed range (at most 0.009) is smaller than the marks.

3 Results and discussion

3.1 Processing settings change peak heights far more than sampling noise does

We first measured how far each setting moves a track away from its original processing. Peak height is the absolute $\log_2$ fold change of the mean count over the top 1% of bins, on peaks that both versions share. Shape is one minus the Spearman correlation between the two versions over 1 kb bins. We compared each change with the resampling floor at the same depth. Peak height moved beyond the floor in 27 of 27 setting levels, by 33 to 763 times the floor (Fig. 2). Depth moved peaks most. At 3.75M reads, peak height changed by 2.5 to 2.8 $\log_2$ units in histone marks and by 3.6 in DNase-seq. The antibody-efficiency proxy at 90% control reads and the 36 bp crop came next, at 1.0 to 1.5 and 1.0 to 1.2 units. Duplicate removal, the mapping-quality filter and run type moved histone peaks by 0.2 to 0.4 units. Shape, in contrast, moved beyond its floor in only 3 of 27 levels.

For model design, this means that a correction must first get the peak level right. At this depth, most settings change shape by less than sampling noise does, so a shape score alone cannot rank them. Each value is a median over the one to three tracks of a mark class and has no seed variation; the floor's seed range is at most 0.009. A reprocessed version shares reads with its original, while the floor compares independent halves. A shape change below the floor is therefore not proof that a setting has no effect. On peaks chosen from the original alone, depth read about 3.0 units in histone marks; the shared peak set removes this selection bias.

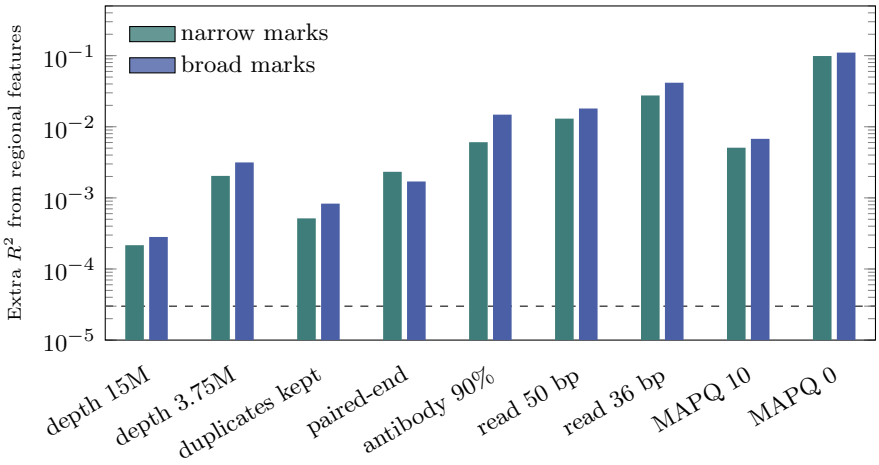


Fig. 3 How much regional features explain of each setting's effect. Each bar is the extra R^2 that GC content, mappability, repeats and peak context add to a signal-level model of the residual left after one genome-wide map (median over the three tracks of a class). The y axis is logarithmic. The dashed line is the largest value of a control with permuted features. There are no error bars, because each value is deterministic.

3.2 Most settings act uniformly along the genome, but the mapping-quality filter and read cropping act mainly in hard-to-map regions

We next asked where along the genome each setting acts. For each setting, we fitted one monotone map from the original to the reprocessed version over the whole genome. We then took the $\log_2$ residual of the reprocessed version in each 1 kb bin. We first explained the residual size by signal level, in 40 quantile bins. We then measured the extra variance explained (R^2) by GC content, mappability, repeat content and peak context together. A setting that acts uniformly leaves no extra R^2, and one genome-wide map then corrects it.

Depth acted uniformly. Its mean absolute residual equalled that of the original thinned to the same depth (0.774 against 0.773 in DNase-seq at 15M), and the features added at most 0.0032 R^2 against 0.0031 for the thinned original. Duplicate removal and run type in histone marks added at most 0.0023. The mapping-quality filter at 0 added 0.099 to 0.110, mostly through mappability (Fig. 3). In the lowest-mappability tenth of bins, its signal was 0.45 to 0.48 $\log_2$ units above what the genome-wide map predicted. The 36 and 50 bp crops added 0.013 to 0.042, through mappability and repeats. A control that permutes the features added at most 0.00003.

For model design, depth, duplicate removal and run type in histone marks need no positional input. Read length and the mapping-quality filter need an input that knows the position, such as a mappability track; Section 3.11 tests this. These readings are exploratory and come from one to three tracks per class.

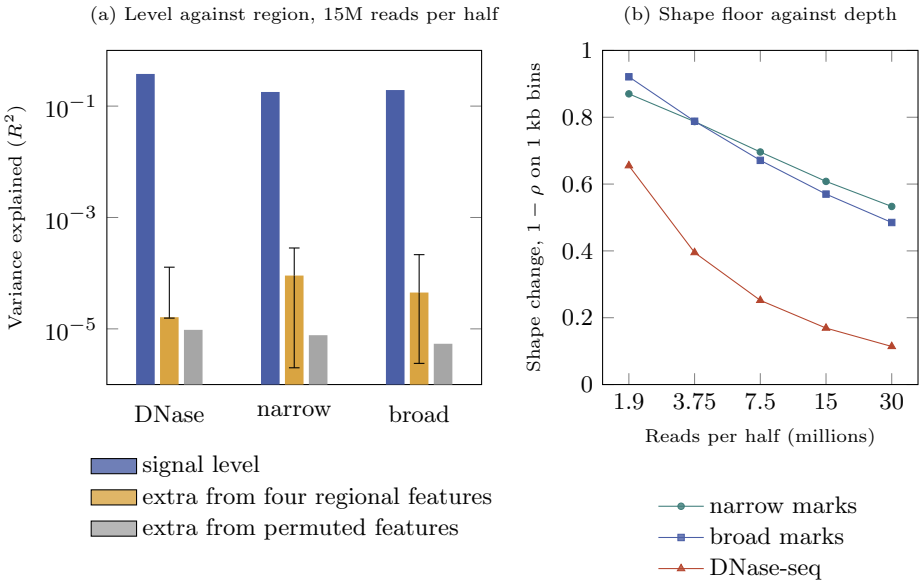


Fig. 4 The resampling floor. (a) Share of the variance of the per-bin disagreement between two random halves that signal level explains, and the extra share that GC content, mappability, repeats and peak context add, or that permuted features add (median over versions; the y axis is logarithmic). Error bars span the smallest to the largest version. (b) Shape change between the two halves as the halves are thinned; each line is the median over the versions of a class. The rightmost point (30M) is extrapolated. The seed range (at most 0.003) is smaller than the marks.

3.3 Sampling noise depends on signal level, not on genomic region

A correction cannot be more accurate than the resampling floor allows, so we mapped that floor. We thinned the two random halves of every version along a depth ladder and compared them bin by bin. At 15M reads per half, signal level explained 0.18 to 0.38 of the variance of the per-bin disagreement. GC content, mappability, repeats and peak context together added at most 0.0003 (Fig. 4a), against 0.00003 for permuted features. The 1 kb shape floor shrank with depth (Fig. 4b). Extrapolated to 30M reads, it was 0.533 for narrow marks, 0.485 for broad marks and 0.114 for DNase-seq. Peak height barely moved under resampling: at most 0.02 $\log_2$ units, even at the lowest depth. Real depth versions matched the thinned halves (median ratio 1.002 over 21 versions).

For model design, a noise model whose variance depends only on the mean, such as the negative binomial, describes this floor; a region-specific noise term is not needed. The 30M values extrapolate a concave curve, so the histone values may be too high. The seed range of the floor is at most 0.003.

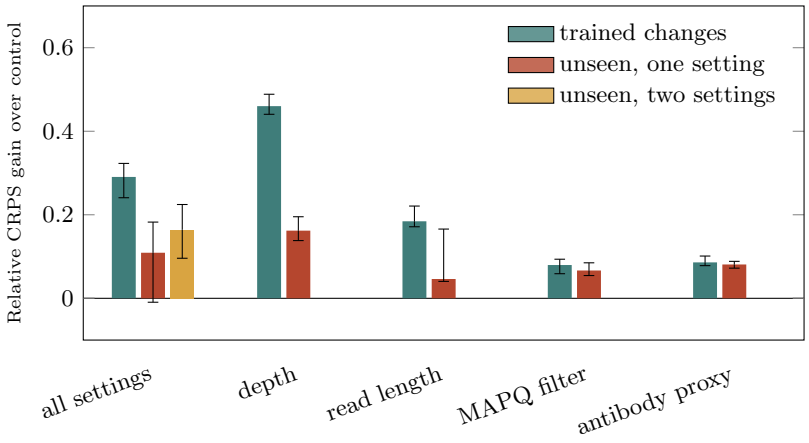


Fig. 5 Benefit of the recorded settings on trained and unseen setting changes, in counts. Each bar is the median, over the 16 model configurations, of the relative CRPS gain of the model over its shuffled-settings control. Error bars span the 25th to the 75th percentile over configurations. Per setting, only changes within one setting exist; duplicate removal and run type have one level and so no unseen change of their own.

3.4 Recorded settings help on trained setting changes, but the benefit shrinks on unseen combinations

We measured the benefit of the settings as the relative CRPS gain of a model over its shuffled-settings control: the control's CRPS minus the model's, divided by the control's. Over the 16 configurations, the median gain on counts was 0.29 on trained setting changes (Fig. 5). It fell to 0.11 on unseen changes within one setting and to 0.16 on unseen changes that combine two settings. For depth, the gain fell from 0.46 to 0.16; for read length, from 0.18 to 0.05. On the p-value track, the gain fell from 0.15 to 0.06 and 0.004.

For model design, the settings clearly carry information that the models use. But the models learn the changes they saw, not a rule that they can apply to new combinations. A model that claims to correct processing must therefore be tested on unseen setting combinations, not only on unseen cell types. These readings are exploratory: no threshold was fixed before the runs.

3.5 The failure to transfer is not a lack of capacity

Depth gives an exact test of transfer, because the real versions obeyed the depth rule within 0.71%. On trained depth changes, every model followed the rule closely: the error was 0.043 to 0.095 over all eight form-and-input combinations and both training scopes (Fig. 6a). On unseen depth changes, the error rose to 0.185 to 0.549. Chaining two trained steps through the original processing brought it back to 0.068 to 0.159. We then tried five fixes on the CNN or the affine form: removing the pseudocount from the log transform, a Poisson loss on the mean, a four times wider conditioning network,

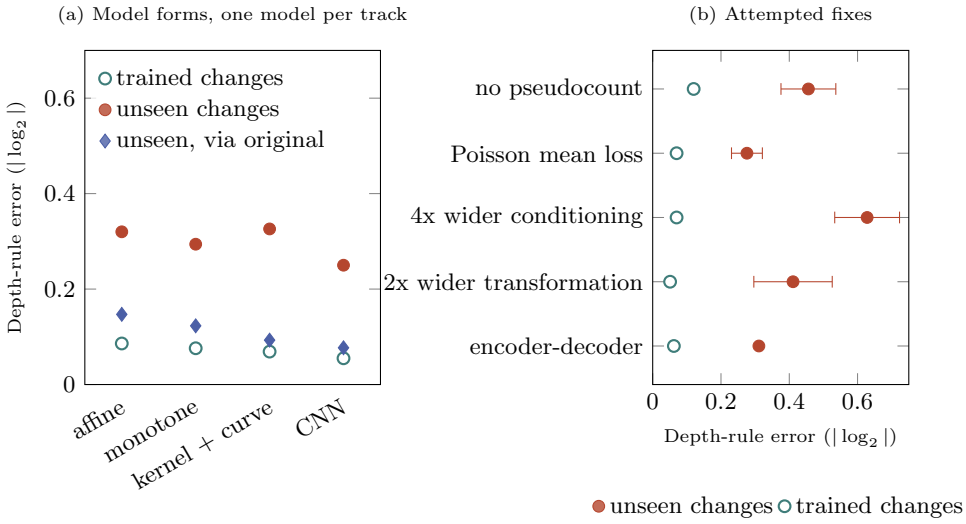


Fig. 6 The depth rule on trained and unseen depth changes. The error is the median absolute $\log_2$ difference between the predicted and the true ratio of total counts; zero means the rule is met. (a) The four transformation forms, each trained as one model per track (median over 3 seeds). Diamonds chain two trained steps through the original processing. (b) Five attempted fixes; error bars on the unseen values span the seed range.

a twice wider transformation and an encoder–decoder design. All of them kept the trained error low (0.05 to 0.12) and left the unseen error at 0.28 to 0.63 (Fig. 6b).

We also tried a closed-form depth offset. It removes the source depth before the transformation and adds the target depth after it. The offset helped only the affine form: its DNase CRPS fell from 3.06 ± 0.113 (1.36 shape $+ 1.70$ level) to 1.247 ± 0.016 ($1.109 + 0.138$), against a threshold of 0.227. The other forms improved significantly in 0 of 36 cells. Even with the offset, no model met the depth rule within 0.10 in any of 64 cells. The shuffled-settings controls of the CNN came much closer (0.054 to 0.135), because the network bends the exact rule when it can read the depth settings.

For model design, the transfer failure is a failure to generalise, not a shortage of capacity. A next model should impose known physical rules, such as depth scaling, rather than learn them, and should check them on unseen changes. These readings are exploratory, and their thresholds were set after the runs.

3.6 A per-bin monotone curve captures most of the correction, and a convolutional network adds little

On trained depth changes, the monotone curve, the kernel-plus-curve form and the CNN closed 0.77 to 0.92 of the gap between the covariate-free monotone baseline and the depth oracle in every mark class (Fig. 7a). Their gain over the baseline was significant in 18 of 18 cells. The affine form did not close the gap (-0.62 to -0.01 in DNase-seq and narrow marks). Over all settings, we compared each form in 12 count cells, one per mark class, training scope and bin set (all bins or the top 1%). The affine form beat its shuffled-settings control in 3 cells, and the other three forms in 11, 11

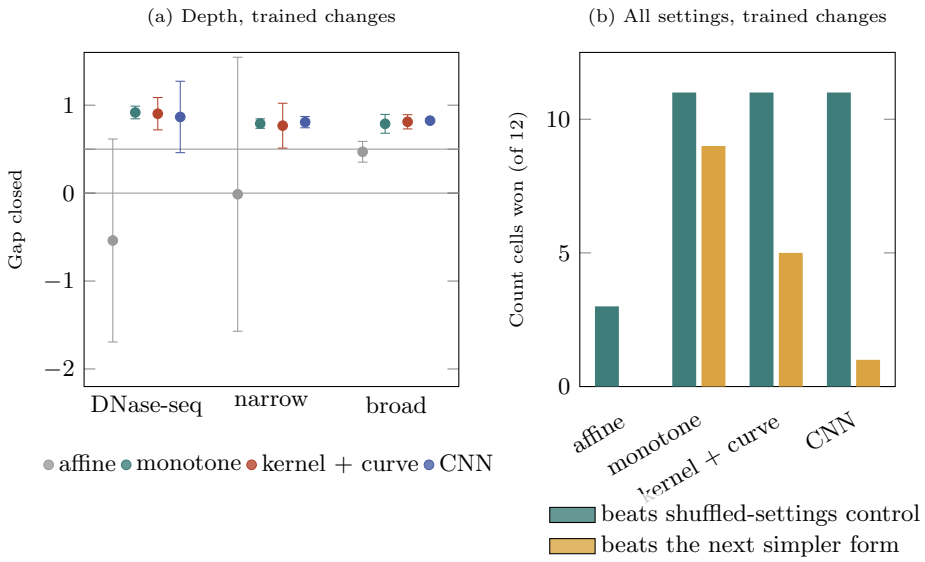


Fig. 7 Capacity of the transformation. (a) Fraction of the gap from the covariate-free monotone baseline to the depth oracle that each form closes, from original to reprocessed version, one model per track; 1 means the oracle is reached. Each error bar extends 2 times the seed range, in gap units, on each side. Grid lines mark zero and 0.5. (b) Number of count cells, out of 12, in which each form beats its shuffled-settings control, and beats the form one step below it, by more than 2 times the seed range.

and 11 (Fig. 7b). Each step up in capacity paid less. The curve beat the affine form in 9 cells, the kernel beat the curve in 5, and the CNN beat the kernel in 1. In DNase-seq, the per-track CRPS was 3.060 ± 0.113 for the affine form (1.360 + 1.700), 0.824 ± 0.004 for the curve (0.816 + 0.007), 0.818 ± 0.027 for the kernel (0.802 + 0.016) and 0.772 ± 0.036 for the CNN (0.760 + 0.012). Most of the affine form's error is in the level part.

For model design, the correction module can be small. A per-bin monotone curve, conditioned on the settings, is a sufficient first design for count data. These readings are exploratory, and the depth oracle was redefined after the run.

3.7 Training on pairs between two reprocessed versions teaches the depth ratio, not the distribution

If the models fail on unseen changes only because they never saw a pair between two reprocessed versions, adding such pairs should fix it. We tested this in a pre-registered experiment with the per-track CNN. We held out the 7.5M depth version entirely and added the two pairs between the 15M and 3.75M versions to training. We scored all pairs that involve the held-out version. Each check had to close at least half (0.5) of the gap between the direct route and the route through the original processing.

The depth rule passed. Its error was 0.148 (seed range 0.123), against 0.195 for the direct route and 0.125 through the original, so the added pairs closed 0.674 of the gap (Fig. 8b). CRPS failed. It was 0.573 (0.466 shape + 0.107 level, seed range

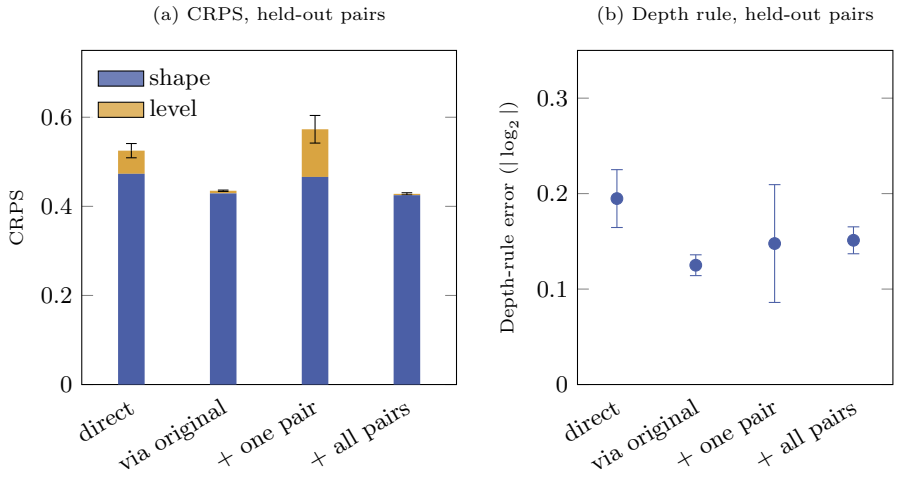


Fig. 8 Adding pairs between reprocessed versions to training (per-track CNN). From left: the direct route, the route through the original processing, training with one added pair of depth versions, and training with all such pairs. (a) CRPS on held-out pairs, split into shape (blue) and level (gold). (b) Depth-rule error. Error bars span the seed range over 3 seeds.

0.062), against 0.525 (0.473 + 0.052) direct and 0.435 (0.429 + 0.006) through the original (Fig. 8a). The gap closed was −0.53, and the gain over the direct route was −0.048 against a required 0.125. The loss sits in the level part. After the test, we also read a version trained on every pair between reprocessed versions that avoids 7.5M. It reached 0.428 (0.425 + 0.003), close to the route through the original.

For model design, a few counterfactual pairs teach the size of a setting's effect but not the full distribution; dense pairs may be needed. DNase-seq drives the CRPS failure, the depth-rule pass varies across seeds (0.089 to 0.213), and the dense-pair result was read after the run on one held-out level.

3.8 Routing through the original processing helps a per-track model but not a model shared across tracks

Another way to reach an unseen change is to chain two trained steps: from the source version to the original processing, then from the original to the target version. For one model per track, this route lowered CRPS on unseen changes from 0.662 (0.606 + 0.057, seed range 0.043) to 0.545 (0.537 + 0.008, 0.008). The gain of +0.118 exceeded the threshold of 0.087 (Fig. 9). The route also met the depth rule (0.077 against 0.250 direct).

We then pre-registered the same test for one model shared across all seven tracks, with 6 seeds. Routing had to win significantly in two thirds of the 98 cells, one cell per pair of source and target setting. It won in 2 and lost in 0. Pooled over cells, routing was lower in 6 of 6 seeds. The pooled gain of +0.027 (direct 0.660 = 0.610 + 0.050; routed 0.633 = 0.576 + 0.057) was still below the threshold of 0.154, because the routed seed range was 0.077. The routed shared model did meet the depth rule (0.089 against a pre-registered limit of 0.16). In an unplanned reading, a shared model

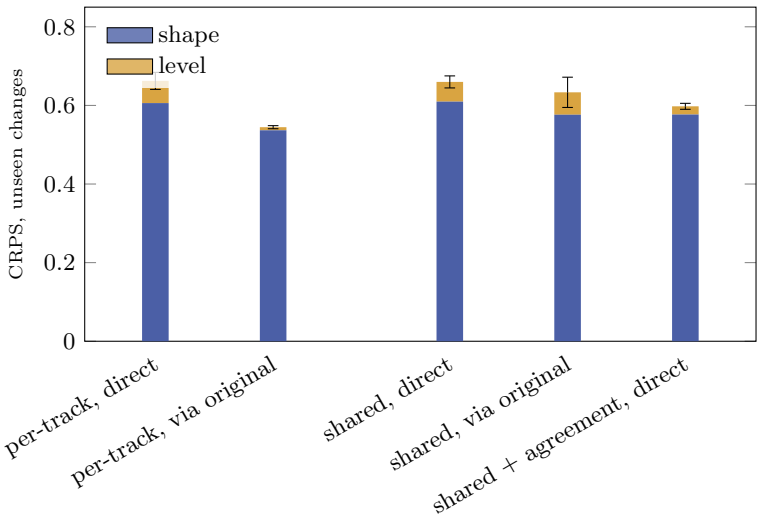


Fig. 9 Routing through the original processing. CRPS on unseen setting changes, split into shape (blue) and level (gold), for one model per track (left pair, 3 seeds) and one model shared across tracks (right three bars, 6 seeds). The last bar is the shared model trained with an agreement loss between the direct and the routed prediction. Error bars span the seed range.

trained with an agreement loss between the two routes scored 0.598 (0.577 + 0.021) on the direct route. Its gain of +0.062 just exceeded the threshold of 0.061.

For model design, routing through a reference version is not a general fix for a shared model, and the agreement loss is only an exploratory lead. The control on trained pairs missed its band on six seeds (−0.006 against 0.006); we counted it on the three seeds shared with the reference model, a decision made after the run. The per-track result was read after its run.

3.9 Predicting counts and deriving *p*-values with the peak caller's formula works almost as well as learning the map

The challenge scored $-\log_{10} p$-value tracks, so a model must also produce them. We compared maps from a count distribution and its matched control to the peak caller's p-value track, first on the true counts of the narrow-mark versions of one track. We scored them by CRPS on $\log(1+p)$, which we call log CRPS. The peak caller's own fragment-window formula scored 0.0865 (0.0864 + 0.0001). The best learned map, the formula plus a small correction network, scored 0.0798 ± 0.0013 (0.0790 + 0.0007). A learned network alone scored 0.0836, and an affine map 0.1791 (Fig. 10).

Fed the CNN's predicted counts, every derived map scored a CRPS of 0.178 to 0.276 on the p-value track. A model that predicts p-values directly scored 0.143. No derived prediction diverged (CRPS above 20), whereas the direct p-value output of the CNN reached a mean DNase CRPS of 1.9×10^4. A next model should therefore predict counts and derive p-values by the formula; the remaining error then belongs to the count model. These results come from one track and one seed of the count model.

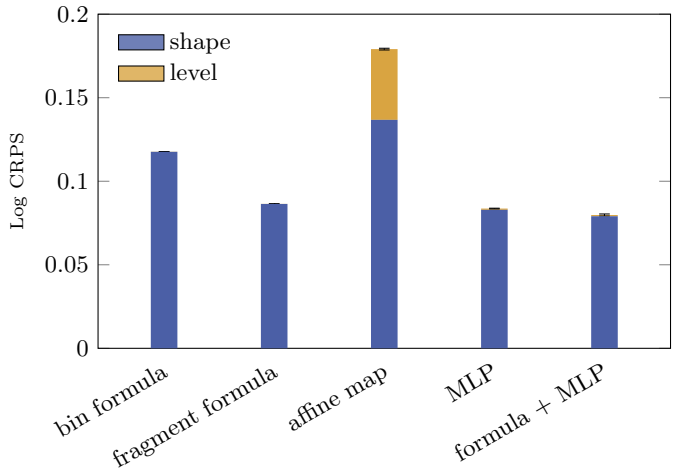


Fig. 10 From true counts to the p-value track (narrow marks, one track). Each bar is the log CRPS of a map, split into shape (blue) and level (gold). The two formulas have no seed; error bars on the learned maps span the seed range over 3 seeds.

3.10 Adding an imputation objective does not change transfer at this scale

CANDI trains its correction jointly with imputation, so the negative results above might not hold there. We pre-registered a test on the shared model. A cross-assay objective predicted another assay of the same cell type from the original version. A denoising objective reconstructed masked bins. At weight 1, cross-assay training moved CRPS on unseen changes by −0.025 (threshold 0.067) and denoising by +0.007 (threshold 0.103). Neither change was significant (Fig. 11). Correction alone scored 0.667 (0.612 + 0.055, seed range 0.019), and cross-assay training 0.642 (0.603 + 0.039, 0.034). At weight 4, cross-assay training came close (−0.034 against 0.039), mostly in the level part (−0.029). Both joint models beat their shuffled-settings controls on unseen changes, by +0.322 and +0.282.

For model design, the earlier negative results on model form probably carry over to joint training at this scale. The trend at weight 4 suggests that a stronger imputation term may improve the level. The power of the test was 41%. Its planned check against the shuffled-settings control on trained pairs could not fail as built, because the control was scored on its own training labels.

3.11 A mappability input improves correction in hard-to-map regions

Section 3.2 showed that read cropping and the mapping-quality filter act through mappability. We pre-registered adding per-bin 36-mer mappability as an extra input channel to the shared model. We scored the lowest-mappability tenth of 1 kb blocks. There, the input lowered CRPS on all 6 of 6 tracks for the 36 bp crop: 0.303 (0.300 + 0.003) against 0.353 (0.322 + 0.031) without it (Fig. 12). It did the same on all 6

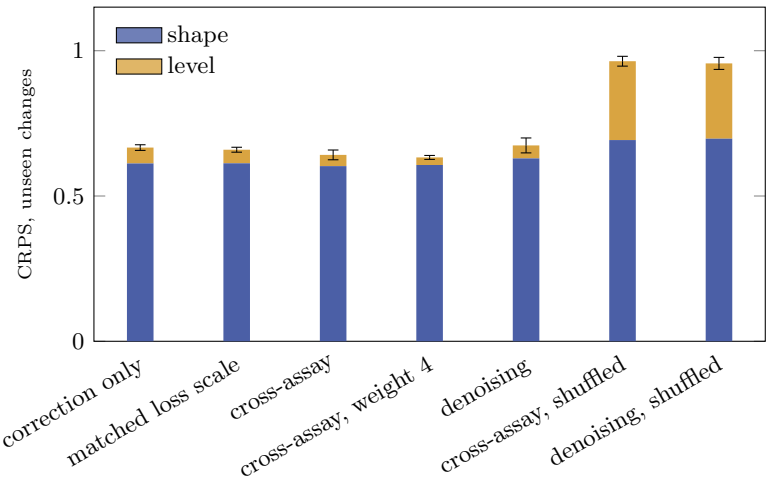


Fig. 11 An added imputation objective, one model shared across tracks. CRPS on unseen setting changes, split into shape (blue) and level (gold), for correction alone, correction with its loss rescaled to match the joint loss, the two joint objectives at weight 1 (cross-assay also at weight 4) and their shuffled-settings controls. Error bars span the seed range over 3 seeds.

tracks for the mapping-quality filter at 0: 0.532 (0.530 + 0.002) against 0.590 (0.569 + 0.020). Mappability shifted by 5 Mb, which keeps the input's distribution but breaks its alignment, did much worse. The aligned input beat it by +0.062 (threshold 0.005) for the filter and by +0.045 (threshold 0.011) for the crop. The gain stayed local: genome-wide CRPS was 0.665 with the input and 0.660 without it. One-hot DNA sequence as input recovered 0.21 to 0.29 of the mappability gain.

For model design, a mappability channel belongs in the first stage of the next model. The evidence is weak, because a pre-registered control failed: CRPS on the depth versions, which should not change, changed by −0.0094 against a seed standard deviation of 0.0011. DNase-seq has no crop version, so six tracks were scored instead of seven. The power of the test was 41%.

3.12 CANDI's current training corpus can teach only depth, and run type has no paired examples

CANDI's two corpora hold processed signal only: 363 challenge experiments and 3,026 merged tracks, all built from released signal files. Random thinning of counts can make depth pairs from them, but it cannot make a pair for any other setting. On the 8-assay panel that CANDI trains on, assay and read length fully determine run type: the conditional entropy of run type is 0.000 bits, against 0.551 bits on the full challenge panel. The model therefore cannot learn run type, the setting behind the challenge's failure, from this panel.

We also probed a trained CANDI model directly. We kept its input fixed and changed only the output-side read length or run type to the value of a reprocessed version, on chr21. Told 36 bp reads, it moved its predicted level by −0.89 to −0.22 $\log_2$ units. The real versions moved by −1.08 to −1.05, mostly because shorter reads

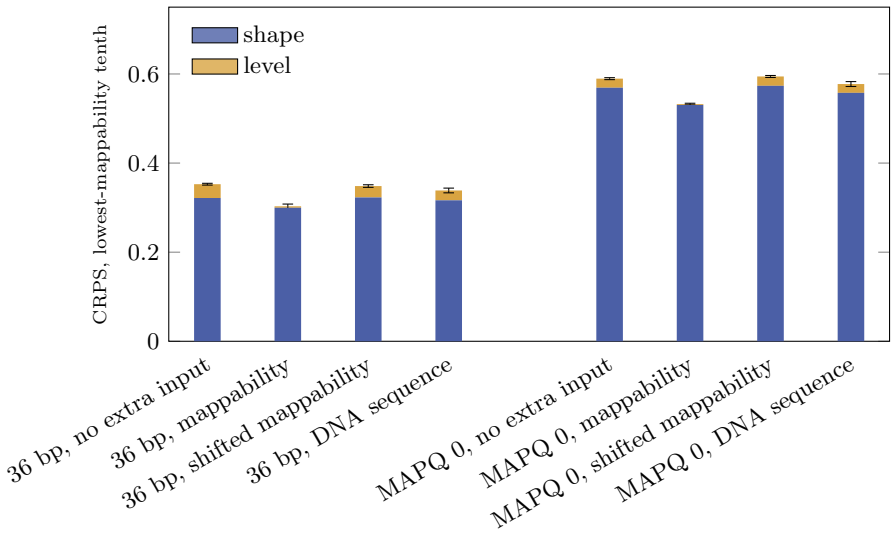


Fig. 12 A mappability input, one model shared across tracks. CRPS in the lowest-mappability tenth of 1 kb blocks on the 36 bp crop (left four) and the mapping-quality filter at 0 (right four), split into shape (blue) and level (gold), as the mean over tracks. Inputs: none, aligned mappability, mappability shifted by 5 Mb, and one-hot DNA sequence. Error bars span the seed range over 3 seeds.

cover fewer bins. Told paired-end, it moved histone predictions by −0.98 to −0.19 (real −0.08 to 0.01) and the DNase prediction by 0.52 to 0.70 (real −0.03; Fig. 13).

For model design, CANDI uses both settings, but not in the direction the real data move. Other settings need new paired data, and read-level settings need the pipeline rerun from raw reads (median 13.4 hours per version). The probe failed its own controls in all three runs: its depth control failed, and its reproducibility test could not separate a real per-bin effect from pure thinning. We therefore read only the level over the whole chromosome; whether CANDI tracks the per-region pattern is unknown.

4 Conclusion

The recorded processing settings carry real information. On setting changes seen in training, models that read them beat a shuffled-settings control, and a per-bin monotone curve captures most of the correction. On unseen combinations of settings, this benefit largely disappears, and no model reproduces the exact depth rule. More capacity, other losses, an added imputation objective and a shared route through the original processing did not change this. The answer to our question is therefore partial: the settings suffice for changes a model has seen, but the models tested here do not learn a correction that transfers. This reading would be settled by a pre-registered transfer test on more tracks.

Several lessons carry over to the next model. It should predict counts with a negative binomial output and derive p-values by the peak caller's formula. Its correction module can be a small monotone curve per bin. It should impose depth scaling as a fixed rule, or at least use the depth rule as a check. It should take a mappability input.

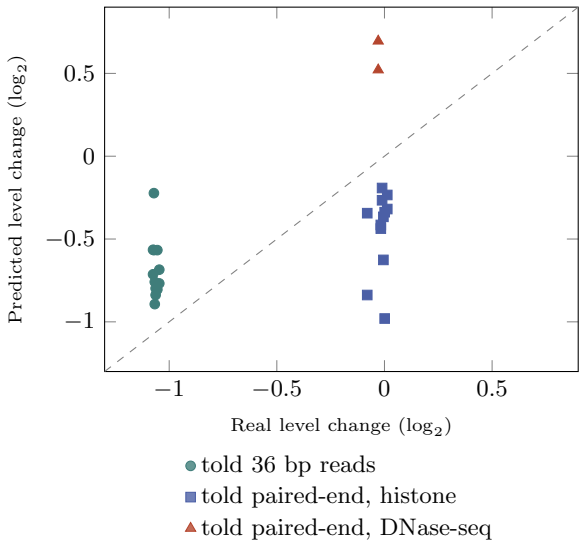


Fig. 13 A trained CANDI model told a reprocessed version's read length or run type. Each point is one track and one of 2 model seeds: the change in predicted level over chr21 against the real change between the original and the reprocessed version. The dashed line marks agreement. There are no error bars; the two seeds of each track are shown as separate points.

It should be judged against a shuffled-settings control, beyond twice the seed range, on unseen setting combinations.

Limitations.

The testbed has seven tracks and three to six seeds per model. Most results are exploratory, with thresholds set after the data were seen. The pre-registered tests had a joint power of only 41% to 51%. The depth oracle was redefined after its run, and no exact reference prediction exists for run type. The p-value maps rest on one track. Finally, the correction was trained on its own, so its failures may not hold inside a full imputation model.

Next steps.

We list the options without ranking them.

- **A first small prototype** with depth and a mappability channel, judged by the shuffled-settings control, the seed range and the depth rule on unseen changes. It costs a testbed-scale run (11 to 36 minutes of training per model). It would teach whether correction survives inside an imputation model.
- **A pre-registered test of the agreement loss**, with 6 seeds. It costs one testbed-scale run. It would teach whether the exploratory lead in Section 3.8 is real.
- **A closed-form depth offset with depth hidden from the network**. It costs a rerun of the offset experiment. It would teach whether imposing the rule removes the depth failure.

- **The run-type effect on more tracks**, from challenge test tracks reprocessed as single-end. It costs CPU-only analysis if suitable products exist. It would teach how large the challenge's failure cause is for each assay.
- **New paired versions for CANDI's corpus.** Read-level settings need the pipeline rerun from raw reads (median 13.4 hours per version; about 1 hour for settings applied after alignment). It would teach whether real pairs fix transfer at scale.

References

[1] Schreiber, J.M., Boix, C.A., Lee, J.W., et al.: The ENCODE Imputation Challenge: a critical assessment of methods for cross-cell type imputation of epigenomic profiles (2023)